UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). All answers are computed deterministically. Worked answers are shown for data representation and assembly; descriptive marking points come from a fixed bank.

2. No placeholders are used; every slot has a worked or descriptive answer.

3. Part A = 10 questions (80 marks); Part B = 1 assembly question (20 marks); total = 100 marks.

4. Computed slots: 4; descriptive slots: 7; placeholders: 0. Questions and answers are for local study use only.

5. Fingerprint based on 20 unique CSC429 papers; output is deterministic for seed 9.

FOR MARKERS' USE ONLY

This marking scheme consists of 9 printed pages.

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PART A — MARKING SCHEME

Worked answers and marking points.

QUESTION A1 (5 marks)

Study Figure 1 and answer the following questions.

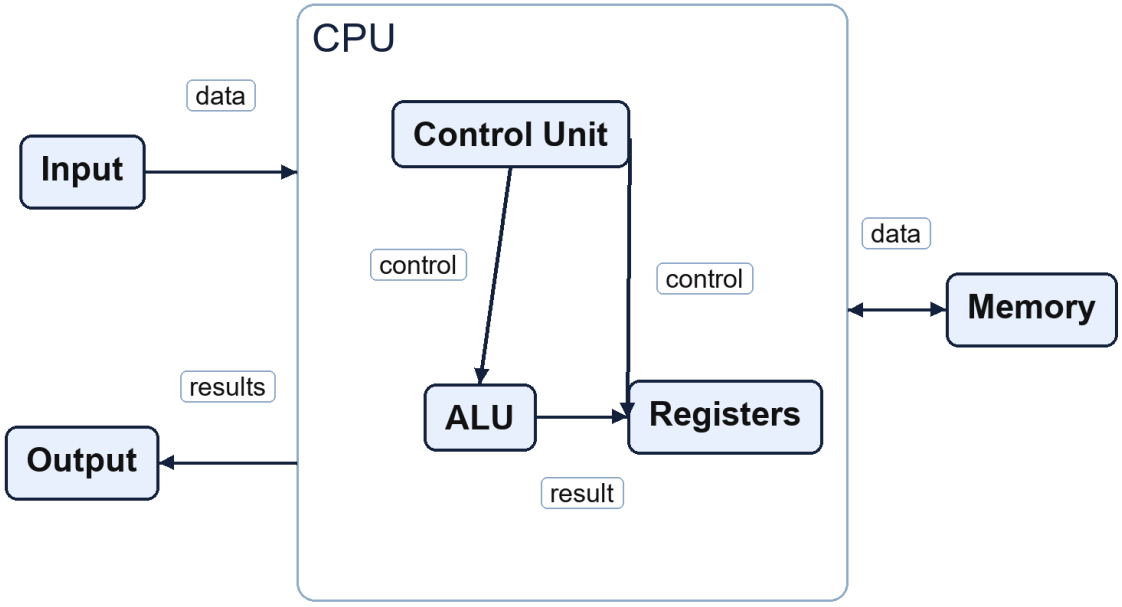


Figure 1: Stored-program computer organization.

(a) (2 marks) Name the computer organization shown in the figure and state its two key principles.

Answer: Von Neumann architecture. Instructions and data are stored in the same memory, and the CPU executes instructions sequentially unless a branch changes the control flow.

(b) (3 marks) Explain the function of the control unit and the arithmetic logic unit in this organization.

Answer: The control unit fetches and decodes instructions and generates the control signals. The ALU performs arithmetic and logical operations and records status flags.

QUESTION A2 (5 marks)

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The program counter and instruction register are important CPU registers.

(a) (2 marks) Differentiate the program counter and the instruction register.

Answer: The PC holds the address of the next instruction; the IR holds the instruction currently being decoded/executed.

(b) (3 marks) Describe the fetch-decode-execute cycle.

Answer: Fetch the instruction at the PC into the IR and increment the PC; decode the opcode and operands; execute the operation and write back the result.

QUESTION A3 **(5 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) (5 marks) Convert -9.5 to IEEE 754 single-precision format. Give the result in hexadecimal and show the sign, exponent and mantissa fields.

Answer: sign=1, exponent=10000010, mantissa=00110000000000000000000 -> C1180000.

QUESTION A4 **(5 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) (2 marks) Represent -18 in 8-bit sign-magnitude form.

Answer: 10010010 (sign bit 1, magnitude 18).

(b) (3 marks) Represent -18 in 8-bit two's complement form.

Answer: 11101110.

QUESTION A5 **(10 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) (2 marks) Convert 237 to binary.

Answer: 237 = 11101101_2.

(b) (2 marks) Convert 10111101_2 to hexadecimal.

Answer: 10111101_2 = BD_{16}.

(c) (3 marks) Convert 11_{16} to octal.

Answer: 11_{16} = 21_8.

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(d) (3 marks) Convert 146_8 to decimal.

Answer: $146_8 = 102_{10}$.

QUESTION A6 **(10 marks)**

Study Figure 2 and answer the following questions.

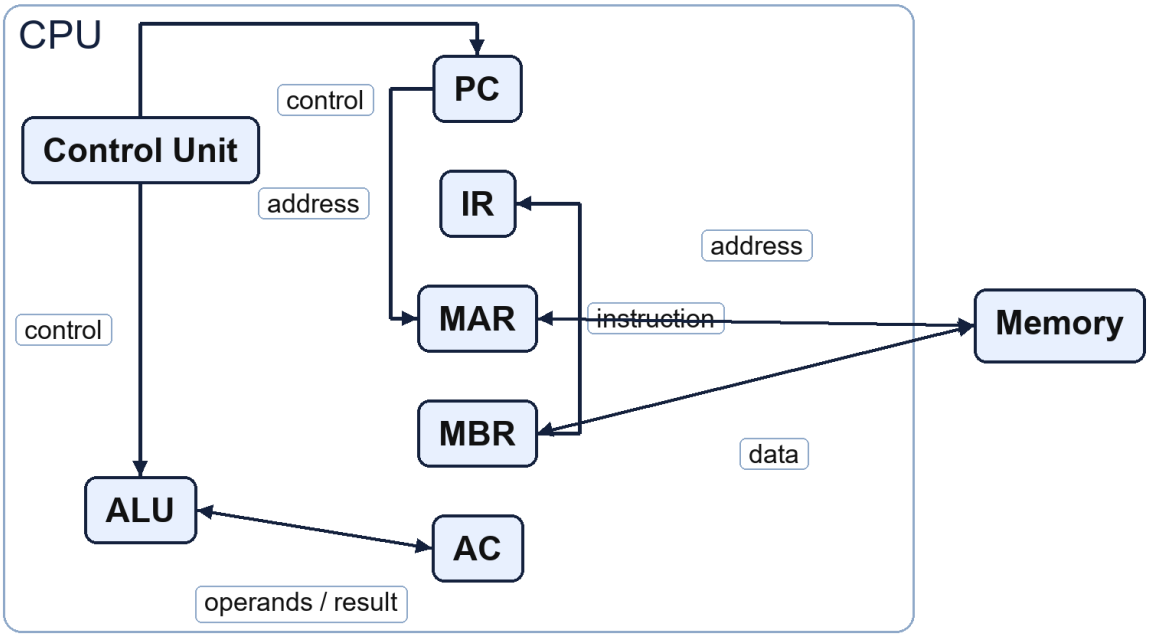


Figure 2: CPU, memory and system bus organization.

(a) (2 marks) Identify the register that holds the address of the next instruction and the register that holds the instruction currently being decoded.

Answer: PC holds the address of the next instruction; IR holds the instruction currently being decoded.

(b) (3 marks) Explain the roles of MAR and MBR during a memory read.

Answer: The MAR holds the address to be accessed. The memory places the selected word in the MBR, which then supplies it to the CPU (for example, to the IR).

(c) (5 marks) Describe the fetch-decode-execute cycle using the components shown in the

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figure.

Answer: Fetch: PC -> MAR; memory read; MBR -> IR; PC is incremented. Decode: the CU interprets the opcode in IR and issues control signals. Execute: operands are read, the ALU performs the operation, the result and flags are written back, and the next instruction is fetched.

QUESTION A7 **(10 marks)**

Buses connect the CPU, memory and I/O devices.

(a) (2 marks) Define a bus.

Answer: A shared set of electrical lines carrying addresses, data and control signals between components.

(b) (4 marks) Compare synchronous and asynchronous bus timing.

Answer: Synchronous uses a common clock and fixed timing; asynchronous uses handshaking, so it is more flexible but has more overhead.

(c) (4 marks) Explain bus arbitration and why it is needed.

Answer: Arbitration decides which master gets the bus when several request it; it prevents conflicting simultaneous transfers.

QUESTION A8 **(10 marks)**

Study Figure 3 and answer the following questions.

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Tag (4 bits)	Line (3 bits)	Offset (9 bits)

Cache state

Line	Valid	Tag	Data
0	0	----	----
1	0	----	----
2	0	----	----
3	1	0010	0xA5
4	0	----	----
5	1	0001	0x3C
6	0	----	----
7	0	----	----

Addresses
0x1A2B
0x2A2B

Figure 3: Direct-mapped cache organization and address fields.

(a) (2 marks) The cache uses a 16-bit address split into a 4-bit tag, a 3-bit line number and a 9-bit offset. For address 0x1A2B, give the tag, line number and offset in binary.

Answer: 0x1A2B = 0001 1010 0010 1011; tag = 0001, line = 101, offset = 000101011.

(b) (4 marks) Using the cache state shown in the figure, determine whether each address is a hit or a miss: (i) 0x1A2B, (ii) 0x2A2B. Explain your reasoning.

Answer: (i) Hit: tag 0001 and line 101; line 5 is valid with tag 0001. (ii) Miss: tag 0010 and line 101; line 5 is valid but its tag is 0001, so the tags do not match.

(c) (4 marks) Explain why the cache compares tags and how temporal and spatial locality improve the hit rate.

Answer: The tag identifies which memory block occupies a line; comparing tags prevents returning data from a different block. Temporal locality means recently used data is likely reused, and spatial locality means nearby addresses are likely used, so keeping blocks in cache raises the hit rate.

QUESTION A9 **(10 marks)**

Interrupts can be maskable or non-maskable.

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(a) (3 marks) Differentiate maskable and non-maskable interrupts.

Answer: Maskable interrupts can be disabled/delayed by the CPU; non-maskable interrupts cannot be ignored and are used for critical events.

(b) (3 marks) Explain what an interrupt handler (ISR) does.

Answer: It saves context, identifies the device/cause, services the request, clears the interrupt and restores context.

(c) (4 marks) Describe TWO methods for handling multiple interrupts.

Answer: Priority-based nesting and polling/sequential service; nested interrupts allow a higher-priority request to pre-empt a lower-priority ISR.

QUESTION A10 **(10 marks)**

Flynn's taxonomy classifies computer architectures by instruction and data streams.

(a) (4 marks) Describe SISD, SIMD, MISD and MIMD.

Answer: SISD: one instruction/one data stream. SIMD: one instruction/ multiple data streams. MISD: multiple instructions/one data stream. MIMD: multiple instructions/multiple data streams.

(b) (2 marks) Give ONE example of SIMD and ONE example of MIMD.

Answer: SIMD: GPU/vector processor. MIMD: multicore CPU or cluster.

(c) (4 marks) Explain the difference between shared-memory and distributed-memory MIMD.

Answer: Shared-memory MIMD uses one address space accessible by all processors; distributed-memory MIMD gives each processor local memory and communicates by messages.

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PART B — MARKING SCHEME

Worked answers and marking points.

QUESTION B1 **(20 marks)**

Write appropriate programs using zero-, one- and three-address instructions to compute Z = (A - B) * C + D / E. Registers A, B, C, D and E must not be modified; X, Y and Z are temporary registers; the final answer must be stored in Z. Available instructions: zero-address PUSH, POP, ADD, SUB, MUL, DIV; one-address LOAD, STORE, ADD, SUB, MUL, DIV; three-address ADD, SUB, MUL, DIV.

(a) (6 marks) Write a zero-address program for the expression.

Answer:

```
PUSH A
PUSH B
SUB
PUSH C
MUL
PUSH D
PUSH E
DIV
ADD
POP Z
```

(b) (6 marks) Write a one-address program for the expression.

Answer:

```
LOAD A
SUB B
STORE X
LOAD X
MUL C
STORE X
LOAD D
DIV E
ADD X
STORE Z
```

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(c) (8 marks) Write a three-address program for the expression.

Answer:
```
SUB X, A, B
MUL X, X, C
DIV Y, D, E
ADD Z, X, Y
```

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